

Q1) Saito's

a) Truth table $\rightarrow 2.5$
 +
 tautology $\rightarrow 0.5$
 NO truth table $\rightarrow 0$

b) Truth table $\rightarrow 1$
 tautology $\rightarrow 0.5$
 modus ponens $\rightarrow 0.5$
 NO truth table $\rightarrow 0$

c) CNF $\rightarrow$

$(P \vee Q \vee R) \wedge (P \vee Q \wedge (P \vee R)) \wedge (R)$
 with steps $\rightarrow 2$

DNF

$(P \wedge S) \vee (P \wedge T) \vee (P \wedge R) \vee (P \wedge T) \vee$
 $(T \wedge Q \wedge R) \vee (T \wedge Q \wedge R)$
 with steps $\rightarrow 2$

Q2 abc: Anurag

a) $X \rightarrow$ original bit
 $Y \rightarrow$ read bit

$$P(X=1|Y=1) = \frac{P(Y=1|X=1)P(X=1)}{P(Y=1)} \quad (0.5)$$

$$P(Y=1) = P(Y=1|X=1)P(X=1) + P(Y=1|X=0)P(X=0) \quad (0.5)$$

$$P(Y=1|X=1) = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} \quad P(Y=1|X=0) = \frac{1}{4}$$

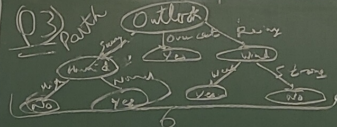
$$P(Y=1) = \frac{3}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{2}$$

Ans $\rightarrow \frac{3}{4}$
 using ans $\rightarrow 0.5$

b) Independent given $M \Rightarrow P(S|H) = P(S|M)P(H|M)$
 $P(M|S) = \frac{P(S|M)P(H|M)}{P(S)} = \frac{P(S|M)P(H|M)}{P(S)} \quad (1)$

c) i) $P(H=1|T=1) = \frac{P(T=1|H=1)P(H=1)}{P(T=1)} \quad (1)$
 ii) Accuracy ≥ 0.5 $\rightarrow 0.5$
 Precision ≥ 0.5 $\rightarrow 0.5$
 Recall ≥ 0.5 $\rightarrow 0.5$

Q2 d) Parth
 Dist Calc $\rightarrow 0$
 $P_{100} \rightarrow 0$
 $X \wedge Y \wedge Z \wedge P \wedge T \rightarrow 0$



1.5 each for 4 of 5's
 Outlook, Temp, Humidity, Wind

Q3) i) Expected ans: for 10 out = Bias = 0, Var = 0.0022, MSE = 0.0022
 for 20 out = Bias = 0, Var = 0.0038, MSE = 0.0038
 if weighted mean method used (final marks 0.9)
 for 10 out = Bias = 0.1, Var = 2.92 = MSE
 for 20 out = Bias = 0.15, Var = 2.93 = MSE

if combined both sets (final marks 1.5)
 Bias = 0, Var = 0.0041 = MSE

if Binomial method used for var $\rightarrow$ full marks
 for 10 out = Bias = 0, Var = 0.01366 = MSE
 for 20 out = Bias = 0, Var = 0.0067 = MSE

if correct ans +1 for Bias +1 for Var +1 for MSE with brief steps showing
 if method correct, but calculation mistake $\rightarrow +0.75$ each
 if partially correct concept $\rightarrow +0.5$
 if only formulae written $\rightarrow +0.25$ each
 +0.25 for finding probability of each (10 & 20 out)
 if weighted mean approach used, or calculation mistakes
 no marks for probability calculations

if correct concept, but finally did not divide by 6
 for combined approach calculated for 3 marks & scaled by 1.5

for weighted means approach $\rightarrow$ final ans $\times 0.9$
 Go through key before approaching TA. If unreasonable queries, you will lose marks

Q4 Memory

a) Support = $1/2$ 0.66 marks
 Confidence = $3/4$ 0.66 marks
 Lift = $9/8$ 0.66 marks

b) Any answer which describes that ϵ greedy enables exploration - 0.5 mark
 Mentioning "exploration" - 0.5 mark

c) $\epsilon/2$ - 0.5 mark
 Any valid reasoning - 0.5 mark

d) Option b - 1 mark

e) 1, 2, 3, 1, 2, 2, 2, 3, 1, 3
 0.6 marks for answer. No marks if stored rather than not been performed in first 3 rounds. Otherwise marks given by case to case bases

7 Krishiv

T/F $\rightarrow 0.5$

Explanation - 1.5

Answer (66) - 1

T/F - 0.5

Risk neutral modelling - 1 (0.5 given if just explained about it)

For explanation & computation - 1.5

T/F - 0.5

Risk averse modelling - 1

Expⁿ and compⁿ - 1.5

(0.52, 0.48) - 1 mark

Final Ans - 1 mark

0.5 - Utility FN

0.5 - Wght bet

Q6 A) 70 100
 $\begin{bmatrix} 2M & Optimal & [70 & 70 & 70 & 70] & [100 & 100 & 100 & 100] \\ 2M & ML & [70 & 70 & 70 & 70] & [100 & 100 & 100 & 100] \end{bmatrix}$

Q6 B) Optimal [40.43, 40.39, 59.02, 70] [61.25, 72.52, 85.72, 100]
 ML [11.07, 15.39, 59.02, 70] [21.43, 27.15, 85.72, 100]

C) 1.5 + 1.5 for calc MSE for both parts

2 Methods of Solving: Assume $[u^*, u^*, u^*, 70]$ & write eqn $\rightarrow$ 4 Equation Solving

Q6 C) Value Iteration from [0, 0, 0, 70]

$\frac{1}{2}$ marks of each correct iteration (at Max for 2 iteration)

A) Optimal $V_1: [0, 0, 5, 6, 70] \frac{1}{2}$
 $V_2: [0, 4, 8, 6, 2, 70] \frac{1}{2}$ and so on
 ML $V_1: [0, 0, 5, 6, 70] \frac{1}{2}$
 $V_2: [0, 4, 8, 6, 2, 70] \frac{1}{2}$

Q6 ABC MANAS

Q8) Jai

a) Only work - 1

Explain - 1

Plg with R - 1

Explain - 1

b) FE - Modify/improve

Scaling - Scale for fixed range

Ext any standard technique

Binning - Continuous data for discrete bins

Q8d) Jai

Dim = 4×13 - 0.5

S₁ node col. - 0.5

0.25/cel

Extra col. - 0.5

Q10) Tathya

Alt 1: Manhattan

Alt 2: h₂ = 2(4-3) + 2(2-1)

Alt 3: h₃ = h₁ (true cost)

or any other heuristic from

(x₁, y₁) or (x₂, y₂) to (x₃, y₃)

For showing $h(n) \leq h^*(n)$

Consistency

Alt 4: even

Alt 5: Manhattan has 13 nodes

Alt 6: h₂ = 2 has 10-12 nodes

Alt 7: h₃ = h₁ has 6 nodes

1 mark for correct no of nodes

2 marks for steps correct g₁ h₁ h₂ values

Q10 C) GBFS has around 6 nodes - 1

Correct h₁ h₂ values in steps - 1

Q10 d) GBFS has 6 while A* has

more nodes expansion but GBFS

doesn't guarantee optimal soln

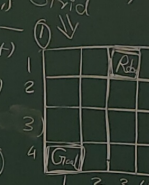
it's greedy algo & A* is optimally efficient

the best algo among all other algo that

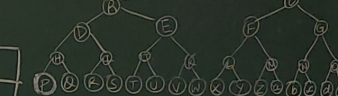
can find optimal soln cuz its consistent &

admissible $\rightarrow$ 0.5

Grid



Q9) Karim



DFS stack: A $\rightarrow$ CB $\rightarrow$ CED $\rightarrow$ CEIH $\rightarrow$ CEIAF

(1 mark) $\Rightarrow$ smmDFS = 5 (1 mark)

BFS queue: A $\rightarrow$ CB $\rightarrow$ ED $\rightarrow$ GFED $\rightarrow$...

(1 mark) $\Rightarrow$ smmBFS = 16 (1 mark)

smmBFS - smmDFS = 16 - 5 = 11 (0.5 marks)

Reasoning (1.5 marks):

The chosen goal node gives maximum value

since smm BFS depends on width of level of

node and smm DFS depends on depth

smmBFS exponential increase in going down

smmDFS linear increase in going down

Goal node any in last level

$\rightarrow$ 0.5